

Python Programming

Unit 05 – Lecture 02: Members, Class Attributes, GC, Abstract Classes

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Repository: <https://github.com/tali7c/Python-Programming>

Quick Links

Core Concepts

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Interactive

Summary

Agenda

1 Core Concepts

2 Demo

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4 Summary

Learning Outcomes

- Use public/private naming conventions in Python classes

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- Inspect built-in class attributes like `__dict__` and `__doc__`

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- Explain garbage collection at a high level

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- Inspect built-in class attributes like `__dict__` and `__doc__`
- Explain garbage collection at a high level
- Create abstract classes using the `abc` module

Public vs Private in Python

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- Python uses conventions (not strict access control)
- `name`: public
- `_name`: internal (“protected” by convention)
- `__name`: name-mangled (harder to access accidentally)

Built-in Class Attributes

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- `__doc__`: documentation string
- `__class__`: object's class
- `__module__`: module name where class is defined

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- Automatically frees memory of unreachable objects
- Python uses reference counting + cycle detection
- Usually you do not manually free memory

Abstract Classes

- Abstract class defines an interface (methods to implement)

```
from abc import ABC, abstractmethod

class Shape(ABC):
    @abstractmethod
    def area(self):
        pass
```

Abstract Classes

- Abstract class defines an interface (methods to implement)
- Prevents creating incomplete objects

```
from abc import ABC, abstractmethod
```

```
class Shape(ABC):  
    @abstractmethod  
    def area(self):  
        pass
```

Demo: Abstract Shape

- File: `demo/abstract_shape_demo.py`

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- Implements:
 - abstract base class `Shape`
 - derived `Circle` and `Rectangle`

Checkpoint 1

Question: What does name mangling mean for an attribute like `--balance`?

Checkpoint 2

Question: Why do we use abstract classes?

Think-Pair-Share

Discuss:

- Create an abstract class `Vehicle` with method `start()`.
- What subclasses can implement it?

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- Built-in attributes help introspection (`__dict__`, `__doc__`)
- GC manages memory automatically
- Abstract classes define required methods for subclasses

Exit Question

Which module is used for abstract base classes in Python?